HAND TRACKING + ROBOTICS

WHAT IF YOUR HAND COULD MOVE A ROBOT?

A webcam turns finger movement into robot commands. No joystick needed.

Imagine moving a hot or hard-to-reach object while the robot does the reaching.



From webcam to motion in four steps

01 / SEE

Webcam

The camera watches one hand move.

02 / FIND

21 landmarks

A model estimates key points on the hand.

03 / MAP

Finger curl

Code turns the points into five curl controls.

04 / MOVE

Robot twin

A 3D hand follows the safe control values.

See the idea come alive

CyberGrip's digital hand picks up a red package, carries it across, and drops it on a platform. A successful delivery reveals the specimen.

→ GRAB / CARRY / DROP

Remote robots

People can control machines from a safer place when a task is hot, risky, or hard to reach.

Hands-free controls

Gesture controls could give people another way to interact with computers and devices.

Try it in simulation

A digital model lets builders test an idea before adding motors and hardware.

CyberGrip is a learning demo. These are possible applications, not results this project proves. The browser pickup is simulated; the optional cardboard build moves five fingers and is not medical or rescue equipment.